

Homework Assignment 9 (Sections 3.1 and 3.2)
Stat 2122 & 3128 - Intro. to Probability & Statistics

Grade: _____/16

Name & Date : _____

1. (P13, page 107) A mail-order computer business has six telephone lines. Let X denote the number of lines in use at a specified time. Suppose the probability mass function of X is as given in the accompanying table. Calculate the probability of each of the following events

x	0	1	2	3	4	5	6
$p(x)$	0.1	0.15	0.20	0.25	0.20	0.06	0.04

(a) {at most three lines are in use} (1 pt)

(b) {fewer than three lines are in use} (1 pt)

(c) {at least three lines are in use} (1 pt)

(d) {between two and five lines, inclusive, are in use} (1 pt)

(f) {at least four lines are not in use} (2 pts)

2. (P23, page 108) A branch of a certain bank in New York City has six ATMs. Let X represent the number of ATMs in use at a particular time of day. The cdf of X is as follows:

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ 0.06, & \text{if } 0 \leq x < 1 \\ 0.19, & \text{if } 1 \leq x < 2 \\ 0.39, & \text{if } 2 \leq x < 3 \\ 0.67, & \text{if } 3 \leq x < 4 \\ 0.92, & \text{if } 4 \leq x < 5 \\ 0.97, & \text{if } 5 \leq x < 6 \\ 1 & \text{if } 6 \leq x \end{cases}$$

Calculate the following probabilities directly from the cdf:

(a) $p(2)$, that is, $P(X = 2)$ (2 pts)

(b) $P(X > 3)$ (2 pts)

(c) $P(2 \leq X \leq 5)$ (2 pts)

(d) $P(2 < X < 5)$ (2 pts)

(e) $P(2 \leq X < 5)$ (2 pts)

Homework Assignment 10 (Section 3.3)
Stat 2122 & 3128 - Intro. to Probability & Statistics

Grade: _____/15

Name & Date : _____

1. (P32, page 116) A certain brand of upright freezer is available in three different rated capacities: 16 ft³, 18 ft³, and 20 ft³. Let X = (the rated capacity of a freezer of this brand sold at a certain store). Suppose that X has the following pmf:

x	16	18	20
$p(x)$	0.2	0.5	0.3

- (a) (4 pts) Compute $E(X)$, $E(X^2)$, $V(X)$, and σ_X

- (b) (2 pts) If the price of a freezer having capacity X is $70X - 650$, what is the expected price paid by the next customer to buy a freezer?

- (c) (2 pts) What is the variance of the price paid by the next customer?

2. (7 pts; P36, page 116) Let X be the damage incurred (in \$) in a certain type of accident during a given year. Possible X values are 0, 1000, 5000, and 10000, with probabilities 0.8, 0.1, 0.08, and 0.02, respectively. A particular company offers a \$500 deductible policy. If the company wishes its expected profit to be \$100, what premium amount should it charge?

Homework Assignment 11 (Section 3.4)
Stat 2122 & 3128 - Intro. to Probability & Statistics

Grade: _____/20

Name & Date : _____

1. (P48, page 123) NBC News reported on May 2, 2013, that 1 in 20 children in the United States have a food allergy of some sort. Consider selecting a random sample of 25 children and let X be the number in the sample who have a food allergy. Then $X \sim \text{Bin}(25, 0.05)$

(a) (2 pts) Determine both $P(X \leq 3)$ and $P(X < 3)$

(b) (2 pts) Determine $P(X \geq 4)$

(c) (2 pts) Determine $P(1 \leq X \leq 3)$

(d) (2 pts) What are $E(X)$ and σ_X ?

(e) (2 pts) In another sample of 50 children, what is the probability that NONE has a food allergy?

2. (5 pts; P55, page 124) Twenty percent of all telephones of a certain type are submitted for service while under warranty. Of these, 60% can be repaired, whereas the other 40% must be replaced with new units. If a company purchases ten of these telephones, what is the probability that exactly two will end up being replaced under warranty? (note that here $n = 10$ and $p = 0.08$)
3. (5 pts; P57, page 124) A certain type of flashlight requires two type-D batteries, and the flashlight will work only if both its batteries have acceptable voltages. Suppose that 90% of all batteries from a certain supplier have acceptable voltages. Among ten randomly selected flashlights, what is the probability that at least nine will work? (note that here $n = 10$ and $p = 0.81$)